

Ebola Data Analytics Portfolio Report

Prepared from the uploaded EbolaVirus-DataSet.zip files. Includes answers to the screenshot questions, charts, selected tables, and Python code used for the analysis.

Uploaded question screenshot

★ BEST QUESTIONS FOR A DATA ANALYTICS PORTFOLIO

If your goal is becoming an entry-level data analyst, these are especially strong:

1. Which countries were most affected by Ebola and why?
2. What factors are most associated with high fatality rates?
3. How have Ebola outbreaks changed over time?
4. Which virus species is most dangerous?
5. Can Ebola deaths be predicted from outbreak characteristics?
6. What transmission factors most influence outbreak spread?
7. Are hospitalization and survival related?
8. What trends emerge before large outbreaks?

Executive answers

1. Which countries were most affected by Ebola and why?

Sierra Leone, Liberia, Guinea, and the Democratic Republic of the Congo carried the largest burden. Sierra Leone had 14,700 cases, Liberia had 11,000, Guinea had 9,600, and DRC had 7,600. Sierra Leone, Liberia, and Guinea were dominated by the large West African outbreak period, while DRC stands out for repeated outbreaks and a high average CFR of 60.5%.

2. What factors are most associated with high fatality rates?

The strongest numeric relationship was lower recovery rate versus higher CFR. Hospitalization rate also had a negative correlation with CFR, about -0.57, suggesting country-years with more hospitalization tended to have lower fatality. This is exploratory and not causal.

3. How have Ebola outbreaks changed over time?

The timeline shifts from smaller, highly fatal early outbreaks to later outbreaks that were much larger in scale. The West Africa period dominates total cases, while some earlier outbreaks had much higher fatality rates.

4. Which virus species is most dangerous?

By weighted CFR in the outbreak file, Sudan virus ranked highest at about 53.2%, followed by Bundibugyo virus at 44.4% and Ebola virus at 43.2%. By total impact, Ebola virus caused the most cases and deaths.

5. Can Ebola deaths be predicted from outbreak characteristics?

A ridge regression using year, country, species, cases, fatality rate, and emergency status produced a leave-one-out R2 of 0.866 and MAE of about 938 deaths. The result is useful as a portfolio demonstration, but deaths are mathematically tied to cases and fatality rate, and the sample is small.

6. What transmission factors most influence outbreak spread?

The highest-risk transmission factors were direct contact with body fluids and unsafe caregiving practices, followed by fruit bat contact, funeral rites with contact, delayed testing, contaminated surfaces, and handling dead primates.

7. Are hospitalization and survival related?

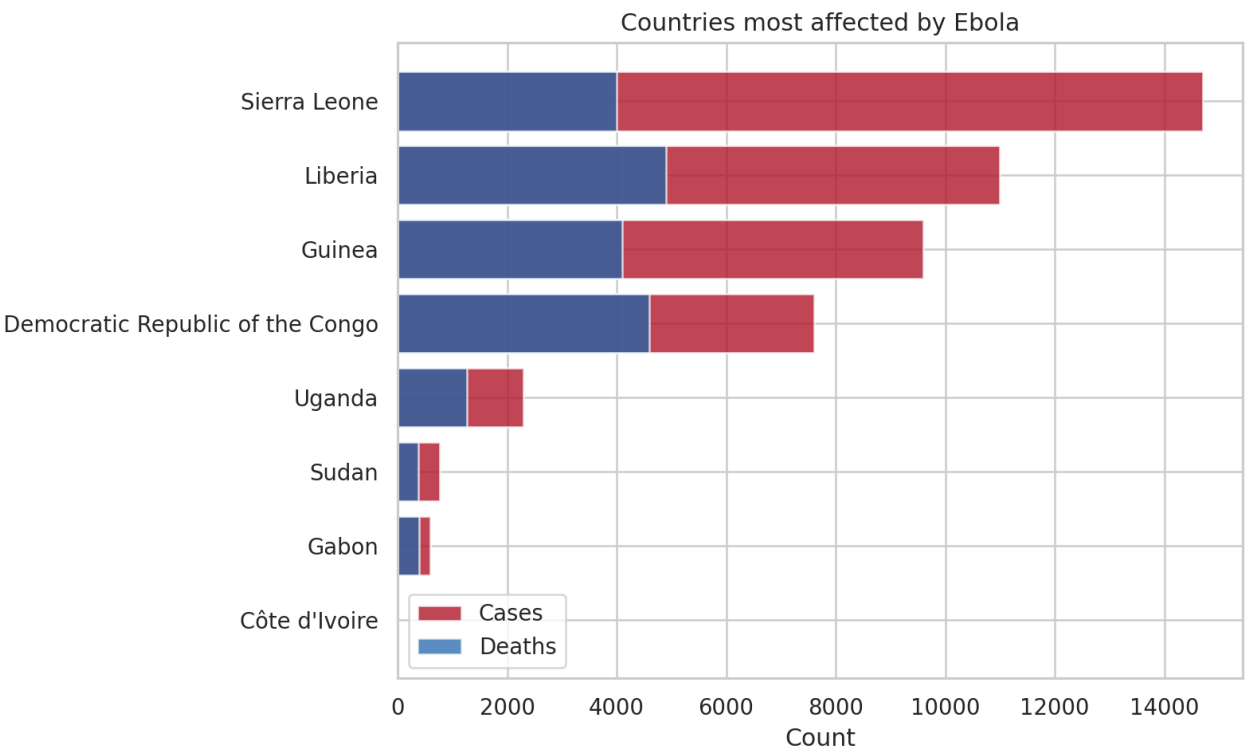
All 10 clinical records were hospitalized, so there is no non-hospitalized comparison group. Among hospitalized patients, 7 recovered and 3 died, for a 70% survival rate in this small clinical sample.

8. What trends emerge before large outbreaks?

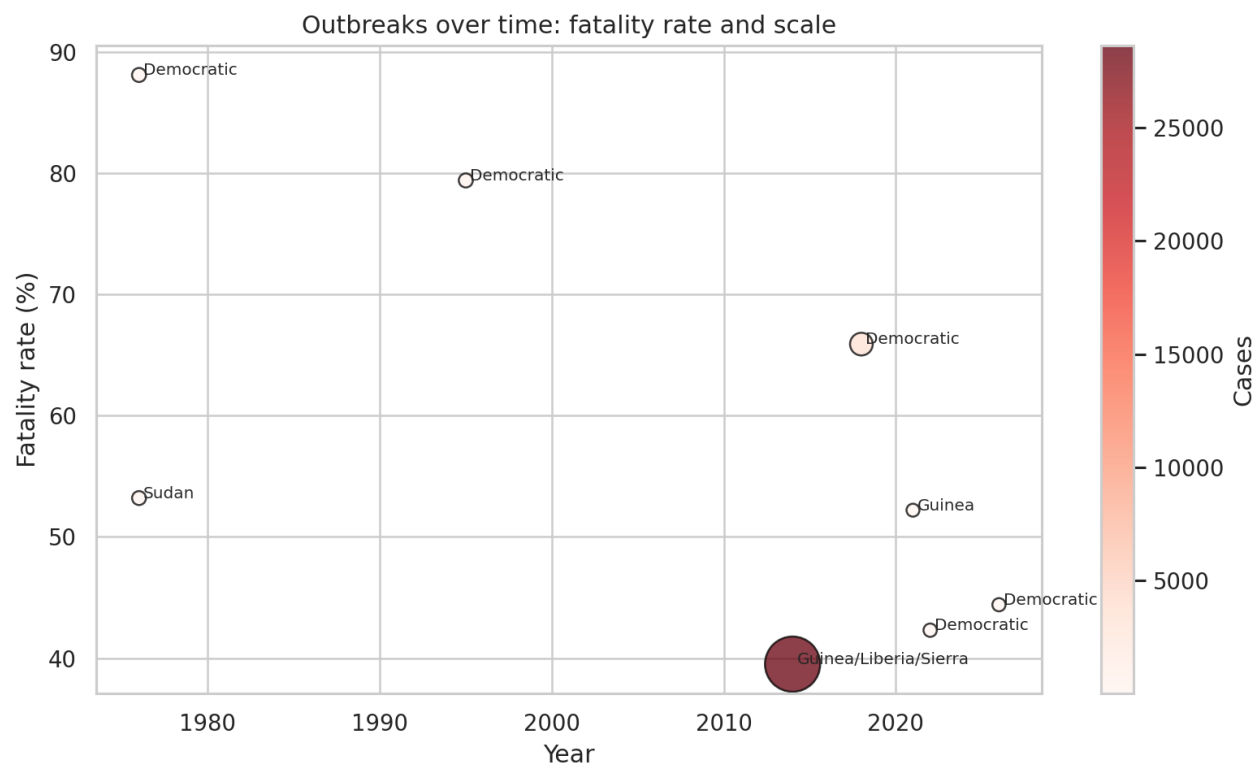
The monthly trend data shows large-outbreak signals around rapid case increases, especially Guinea in December 2014, Sierra Leone in January 2015, and Liberia in September 2015. The key red flags are rising cases, ongoing deaths, delayed testing, and high-contact transmission settings.

Key table: country impact

country	total_outbreaks	total_cases	total_deaths	average_cfr
Sierra Leone	3	14700	4000	27.2
Liberia	2	11000	4900	44.5
Guinea	5	9600	4100	42.7
Democratic Republic of the Congo	11	7600	4600	60.5
Uganda	7	2300	1260	54.8
Sudan	2	770	371	48.2
Gabon	5	600	390	65.0
Côte d'Ivoire	1	1	0	0.0

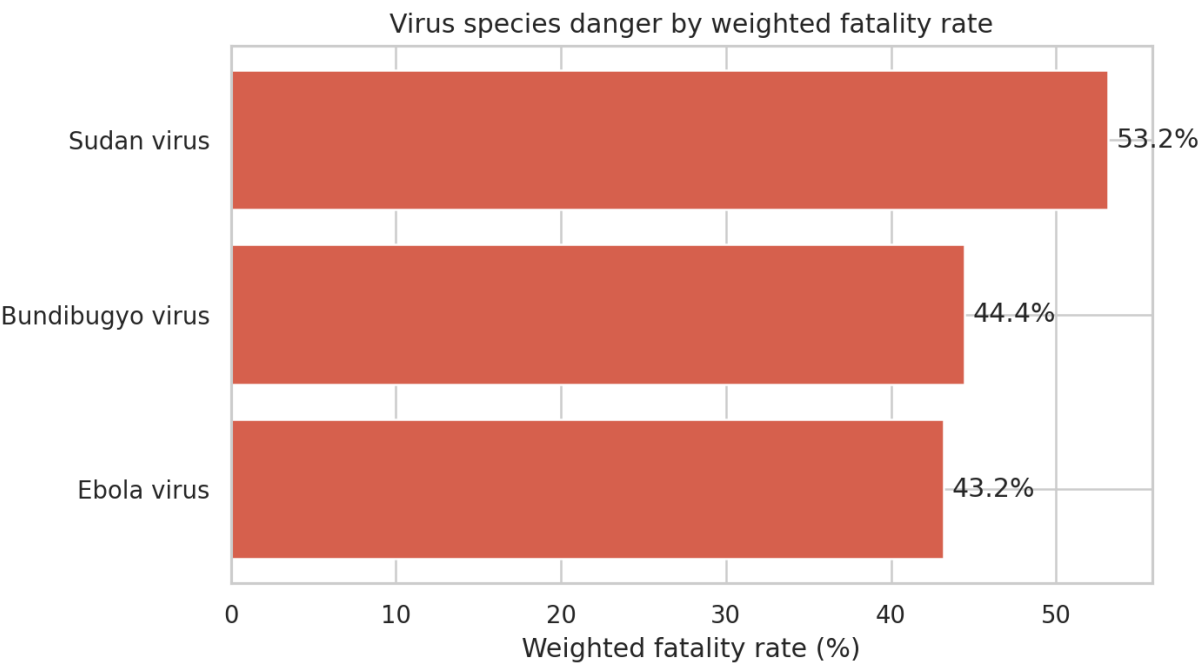


Outbreaks over time



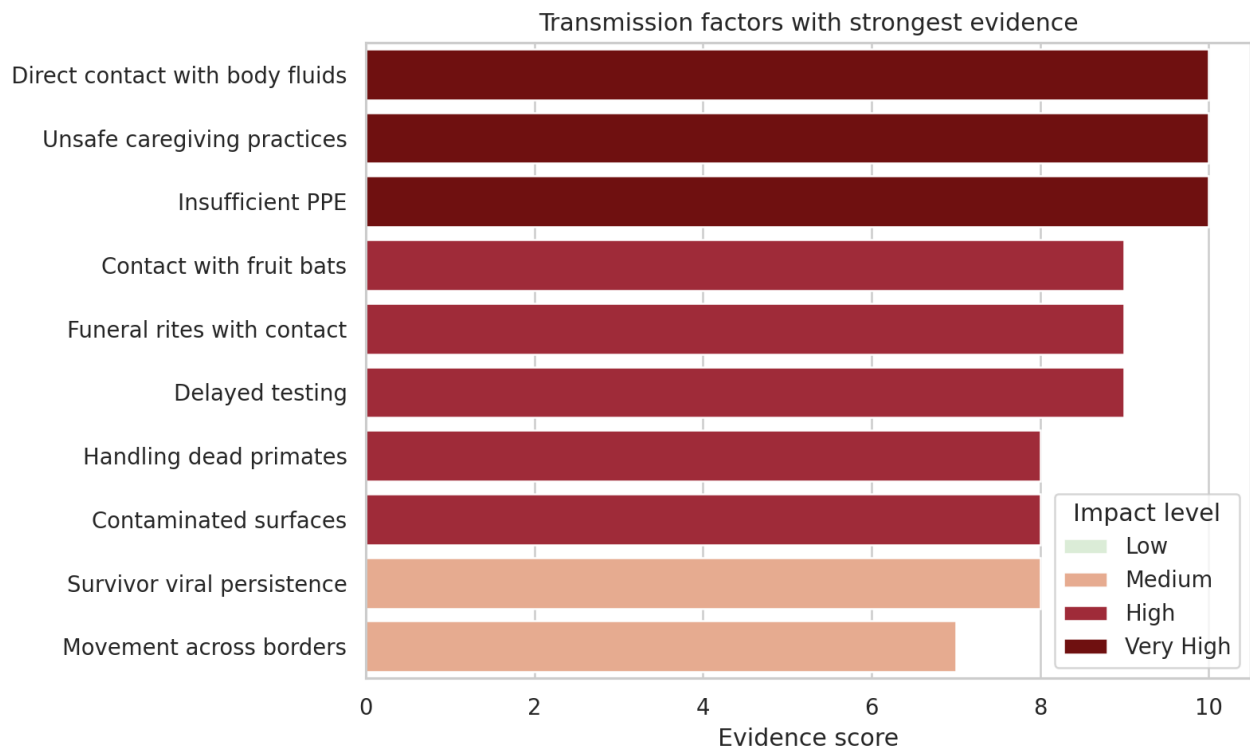
Virus species danger

virus_species	outbreaks	total_cases	total_deaths	weighted_cfr
Sudan virus	1	284	151	53.17
Bundibugyo virus	1	90	40	44.44
Ebola virus	6	32872	14194	43.18



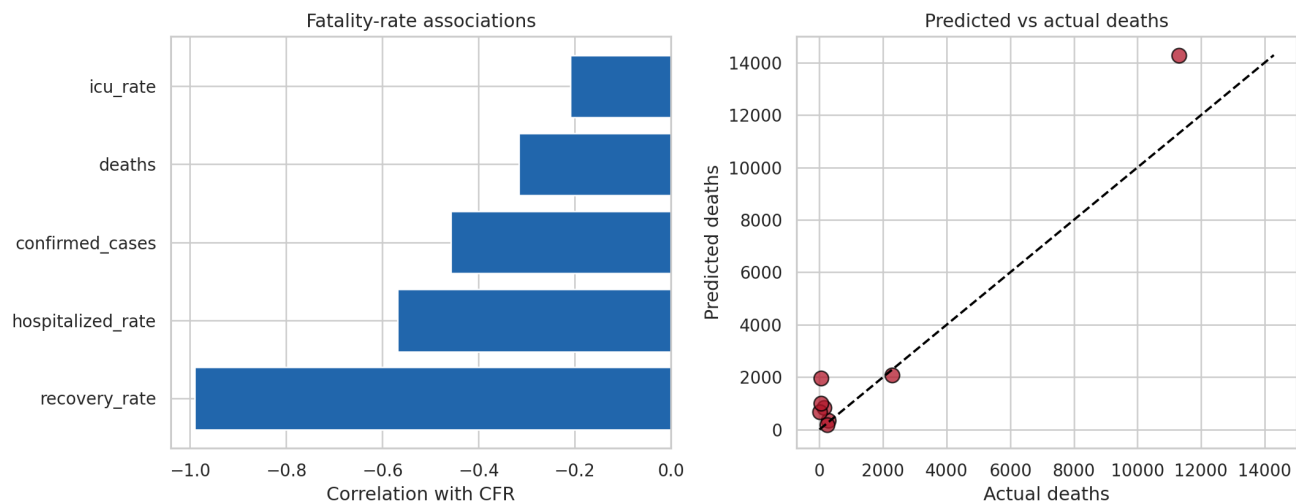
Transmission factors

factor_category	factor	impact_level	evidence_score
Human-to-human	Direct contact with body fluids	Very High	10
Human-to-human	Unsafe caregiving practices	Very High	10
Healthcare	Insufficient PPE	Very High	10
Wildlife	Contact with fruit bats	High	9
Human-to-human	Funeral rites with contact	High	9
Diagnostics	Delayed testing	High	9
Wildlife	Handling dead primates	High	8
Healthcare	Contaminated surfaces	High	8
Recovery	Survivor viral persistence	Medium	8
Travel	Movement across borders	Medium	7



Fatality associations and prediction model

The model results were leave-one-out $R^2 = 0.866$ and $MAE = 937.5$ deaths. This is directionally useful, but the dataset is very small and deaths are closely tied to cases and fatality rate.



Hospitalization and survival summary

hospitalized	outcome	patients
1	Deceased	3
1	Recovered	7

All 10 clinical records were hospitalized, so a clean hospitalized-vs-not-hospitalized comparison is not available. Seven recovered and three died.

Code appendix

Code block 1

```
# Load uploaded Ebola datasets
import zipfile, pandas as pd, numpy as np
from pathlib import Path
from tqdm.auto import tqdm
zip_path = Path('EbolaVirus-DataSet.zip')
extract_dir = Path('ebola_data')
with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    for member in tqdm(zip_ref.namelist()):
        zip_ref.extract(member, extract_dir)
dataframes = {p.stem: pd.read_csv(p) for p in extract_dir.glob('*.csv')}
```

Code block 2

```
# Clean and engineer core fields
country_yearly_df = dataframes['ebola_country_yearly'].copy()
master_df = dataframes['ebola_master'].copy()
outbreaks_df = dataframes['ebola_outbreaks'].copy()
clinical_df = dataframes['ebola_clinical'].copy()
for frame in [country_yearly_df, master_df, outbreaks_df, clinical_df]:
    for col in frame.columns:
        if col not in ['country', 'iso3', 'who_region', 'virus_species', 'outcome']:
            frame[col] = pd.to_numeric(frame[col], errors='ignore')
outbreaks_df['start_date'] = pd.to_datetime(outbreaks_df['start_date'])
outbreaks_df['year'] = outbreaks_df['start_date'].dt.year
clinical_df['survived'] = clinical_df['outcome'].str.lower().isin(['survived', 'recovered']).astype(int)
clinical_df['died'] = clinical_df['outcome'].str.lower().isin(['died', 'deceased']).astype(int)
```

Code block 3

```
# Country impact and species danger
country_impact = master_df.sort_values('total_cases', ascending=False)
species_impact = outbreaks_df.groupby('virus_species', as_index=False).agg(
    outbreaks=('country', 'count'),
    total_cases=('cases', 'sum'),
    total_deaths=('deaths', 'sum'),
    weighted_cfr=('deaths', lambda vals: 100 * vals.sum() / outbreaks_df.loc[vals.index, 'cases'].sum())
).sort_values('weighted_cfr', ascending=False)
```

Code block 4

```
# Fatality associations and prediction model
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import Ridge
from sklearn.model_selection import LeaveOneOut, cross_val_predict
from sklearn.metrics import r2_score, mean_absolute_error
analysis_yearly = country_yearly_df.copy()
analysis_yearly['hospitalized_rate'] = analysis_yearly['hospitalized'] / analysis_yearly['confirmed_cases']
analysis_yearly['icu_rate'] = analysis_yearly['icu_admissions'] / analysis_yearly['confirmed_cases']
analysis_yearly['recovery_rate'] = analysis_yearly['recovered'] / analysis_yearly['confirmed_cases']
fatality_corr = analysis_yearly.corr(numeric_only=True)['case_fatality_rate'].sort_values(key=abs, ascending=False)
features = ['year', 'country', 'virus_species', 'cases', 'fatality_rate', 'who_emergency_status']
model_df = outbreaks_df[features + ['deaths']].dropna()
preprocess = ColumnTransformer([
    ('num', StandardScaler(), ['year', 'cases', 'fatality_rate']),
    ('cat', OneHotEncoder(handle_unknown='ignore'), ['country', 'virus_species', 'who_emergency_status'])
])
model = Pipeline([('preprocess', preprocess), ('ridge', Ridge(alpha=1.0))])
pred_vals = cross_val_predict(model, model_df[features], model_df['deaths'], cv=LeaveOneOut())
print(r2_score(model_df['deaths'], pred_vals))
print(mean_absolute_error(model_df['deaths'], pred_vals))
```

Code block 5

```
# Example visualization pattern
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style='whitegrid')
fig, ax = plt.subplots(figsize=(9, 5.5))
top_countries = country_impact.sort_values('total_cases')
```



```
ax.barh(top_countries['country'], top_countries['total_cases'], label='Cases')
ax.barh(top_countries['country'], top_countries['total_deaths'], label='Deaths')
ax.set_title('Countries most affected by Ebola')
ax.legend()
plt.tight_layout()
plt.show()
```